JAYA HAREESWAR REDDY

Generative AI Engineer | Machine Learning & Deep Learning

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Professional Summary

Generative AI Engineer experienced in developing LLM-powered applications, retrieval-augmented generation (RAG) pipelines, and tool-calling agent workflows. Skilled in Python, FastAPI, prompt engineering, embeddings, vector search, structured outputs, and model integration. Knowledge of machine learning, deep learning, data analysis, and LoRA/QLoRA, with research experience in CNN–VGG16-based medical image classification.

Professional Experience

AI Engineer | Taskwin GmbH *Apr 2026 – Present*

- Develop LLM-powered backend services using prompt engineering, structured outputs, and robust error handling.
- Build RAG pipelines involving document chunking, embeddings, and vector search for context-aware retrieval.
- Implement tool-calling and multi-step agent workflows integrated with backend tools and services.
- Validate model outputs and handle failures, latency monitoring, and token-usage tracking in AI workflows.

Research Experience

Brain Tumor Detection Using CNN–VGG16 Hybrid *Jan 2025 – May 2025*
Python, TensorFlow, Pandas, Scikit-learn, CNN, VGG16

- Developed a hybrid CNN–VGG16 image-classification model to categorize brain MRI images across 8 classes.
- Prepared and processed image datasets using Pandas and implemented the deep-learning workflow with TensorFlow.
- Evaluated model performance using accuracy, precision, recall, F1-score, and a confusion matrix to assess classification performance across classes.
- Achieved 99% classification accuracy on the evaluated dataset.

Technical Skills

Programming: Python, SQL, JavaScript, Java, C++
Generative AI: Prompt Engineering, RAG, Embeddings, Vector Search, Tool Calling, Structured Outputs, LoRA, QLoRA
ML/DL: TensorFlow, Scikit-learn, CNN, VGG16, NLP, Named Entity Recognition, Semantic Matching
Data Engineering: Pandas, NumPy, Apache Kafka, Apache Airflow
Backend/Databases: FastAPI, REST APIs, MongoDB, MySQL, Supabase, Milvus
Tools & DevOps: Docker, Kubernetes, Selenium, Postman, VS Code, Git, GitHub

Projects

Prompt-to-UI Generator
Python, FastAPI, OpenAI API, React.js, Tailwind CSS

- Built an LLM application that generates React interfaces from natural-language prompts, with live previews, streaming, syntax-error handling, and iterative refinement.
- Integrated the FastAPI backend with the React frontend to deliver generated UI components and live previews.

AI Resume Analyzer & Job Matcher
Python, FastAPI, spaCy, Selenium, MongoDB

- Developed NLP-based resume extraction and job matching; scraped and ranked 1,000+ LinkedIn listings by role and skill relevance.
- Automated resume and job data processing to identify relevant skills and improve candidate-job matching.

FutureTrackAI
Python, FastAPI, OpenAI API, MongoDB, React.js

- Built an LLM-powered career recommendation system with structured outputs, JSON-schema validation, retries, and 12 FastAPI endpoints.
- Integrated AI-generated career recommendations with MongoDB and REST APIs to support personalized guidance.

Education

SRM University–AP *2022 – 2026*
B.Tech, Computer Science and Engineering | CGPA: 8.1/10
Coursework: Machine Learning, Deep Learning, Probability and Statistics, Linear Algebra

ARKA Junior College *2020 – 2022*
Intermediate, MPC | 91%

Certifications

Claude Code in Action – Anthropic
MongoDB Associate Developer